

GTU CSE 241 / 501 Quiz

Std Code:1010

```
#ifndef shapec-h
#define shapec-h
#include <iostream>
#include "Shape.h"
using namespace std;
#include <vector>
template <class T>
#include <typeinfo>
class ShapeContainer: public Shape<T>{
private:
vector<T*> data;
int used;

public:
ShapeContainer(int Used=0): used(Used){}
ShapeContainer(const ShapeContainer& o){
: used(o.used){}
for(int i=0; i<used; i++){
data[i] = o.data[i];
}
void add(T& other){
data[used++] = &other;
}
double total() const{
double sum=0.0;
for(int i=0; i<used; i++){
sum += area();
}
return sum;
}
int numCircle() const{
int counter=0;
for(int i=0; i<num; i++){
if(typeid(*data[i]) == typeid(Circle))
++counter;
}
return counter;
}
void printInfo() override{
cout << "Total Area: " << total() << endl;
}
}; // close class
#endif
```

```
#include "Shape.h"
#include "Container.h"
```

```
int main() {
```

```
ShapeContainer<Shape> shape1;
```

```
Circle circle1;
```

```
Rectangle rect1;
```

```
shape1.add(circle1);
```

```
shape1.add(rect1);
```

```
shape1.printInfo(); // To show total
```

```
int x=numCircle(); // number of circle
```

```
cout<<"Total Circle counter:"<<x<<endl;
```

```
return 0;
```

```
}
```